Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: General Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Force and Pressure: Why Sharp Things Cut Better

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Unit 1: Force and Pressure: Why Sharp Things Cut Better

Learning Objectives

• Students will define pressure as force acting on a unit area.

• Students will explain why a smaller contact area produces greater pressure.

• Students will give everyday examples of pressure in action.

Key Concepts

• Pressure equals force divided by area: P = F / A, measured in pascals.

• The same force spread over a large area gives low pressure; concentrated on a small area it gives high pressure.

• That is why a sharp knife cuts easily while a blunt one does not, and why a person sinks less on wide snowshoes than in boots.

• High pressure can be useful, like a needle piercing cloth, or harmful, like a heavy load cracking a weak surface.

Teaching Methodology (Step by Step)

1. Press a fingertip against a desk, then press a pencil point with the same effort, and ask why one hurts more.

2. Introduce P = F / A and work out two simple comparisons on the board.

3. Compare a sharp knife and a blunt knife, and a camel's wide feet with a horse's narrow hooves.

4. Students predict which of two objects sinks deeper in soft soil and explain why.

5. Students write one example where high pressure helps and one where low pressure helps.

Classroom Activities

• Push-pin test: students press pins with flat and pointed ends into soft board and compare depth.

• Shoe sort: groups sort pictures into high-pressure and low-pressure examples.

• Pressure hunt: students list five pressure examples from home and school.

Assessment Questions

1. Define pressure and write its formula.

2. Why does a sharp needle pierce cloth more easily than a blunt rod?

3. A force of 20 N acts on 2 m^2. What is the pressure?

Homework

• Explain why tractors have wide tyres, using the idea of pressure.

• Find two tools at home that use high pressure and two that use low pressure.

Time Allocation

40 minutes